

Flow Meter MAPS Template (Instrumentation)

The Flow Meter MAPS template is designed to capture the data from a Flow meter and calculate the different flow rates.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is a flow meter, which provides no control over the flow meter it only reads flow meter data.

This document describes the Advanced and Basic models of the Flow Meter MAPS Templates. For details of their required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Flow Meter Functional Philosophy

The Flow meter block receives an Analogue signal which represents the Instantaneous flow in meters per second. This signal is then scaled to give the instantaneous flow Meters Cube per minute, total flow Meters Cube, total flow hours, daily total flow Meters Cube, and daily total flow Hours etc.

Field Interlocks

- None

Interlock fault recovery process – None

The start of flow calculation is enabled by setting the minimum flow rate on the **Setup** screen of the flow meter.

Typical Faceplate Graphics for the Flow Meter MAPS Template

FM Top		FM Bottom		FM Left		FM Right		
0000_ABCDE_123  L/S		0000_ABCDE_123  L/S		0000_ABCDE_123  L/S		0000_ABCDE_123  L/S		
0000_ABCDE_123 L/S		Flow Meter Control Name						
		Instantaneous flow in meters per second						
Graphics legend:								
Line colour	Fill colour	DOL status						
Black	Grey	No Flow						
Black	Green	Flowing						
Black	Yellow/Red	Fault Active						

Signal Description		Agent Type	Advanced	Basic
SCADA Control Words:				
SSCL 0	Control Word	Marshal	x	x
SSCL 1	Minimum IN (Raw Minimum)	Analog	x	x
SSCL 2	Maximum IN (Raw Maximum)	Analog	x	x
SSCL 3	Minimum OUT (Eng Minimum)	Analog	x	x
SSCL 4	Maximum OUT (Eng Maximum)	Analog	x	x
SSCL 5	Minimum Flow	Analog	x	x
SSCL 6	High Flow Set point	Analog	x	
SSCL 7	Low Flow Set point	Analog	x	

Signal Description		Agent Type	Advanced	Basic
SCADA Status Words:		Analog	x	x
SSSL 0	Instantaneous Flow Litres/Second	Analog	x	x
SSSL 1	Instantaneous Flow Metres Cubed/Hour	Analog	x	x
SSSL 2	Totalised Flow Value Metres Cubed (M3)	Analog	x	x
SSSL 3	Totalised Flow Hours	Analog	x	x
SSSL 4	Current Daily Total Flow (M3)	Analog	x	x
SSSL 5	Current Daily Flow Hours (Hrs.)	Analog	x	x
SSSL 6	Resettable Flow Value (M3)	Analog	x	x
SSSL 7	Resettable Flow Hours Value (Hrs.)	Analog	x	x
SSSL 8	Kilowatt-hour vs. Flow	Analog	x	x
SSSL 9	Status Word	Marshal	x	x

Digital Inputs:		Advanced	Basic
DI 0	Flow Meter Pulse	x	x
Digital Outputs:			
Analog Inputs:			
AI 0	Flow Meter Flow Rate	x	x

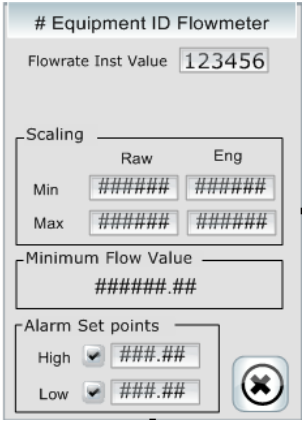
SCADA Control Word (Adroit Marshal Agent)		Event Logged	Advanced	Basic
Bit 0	Reset Flow	x	x	x
Bit 1	Reset Hours	x	x	x
Bit 2	Reset Daily	x	x	x
Bit 3	Spare			
Bit 4	Spare			
Bit 5	Spare			
Bit 6	Spare			
Bit 7	Spare			
Bit 8	Spare			
Bit 9	Spare			
Bit 10	Spare			
Bit 11	Spare			
Bit 12	Spare			
Bit 13	Spare			
Bit 14	Spare			
Bit 15	Spare			
SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Basic
Bit 0	Daily Reset		x	x
Bit 1	Flowing		x	x
Bit 2	Spare		x	x
Bit 3	High Flow Alarm	x	x	
Bit 4	Low Flow Alarm	x	x	
Bit 5	Spare			
Bit 6	Spare			
Bit 7	Spare			
Bit 8	Spare			
Bit 9	Spare			
Bit 10	Spare			
Bit 11	Spare			
Bit 12	Spare			
Bit 13	Spare			
Bit 14	Spare			
Bit 15	Spare			

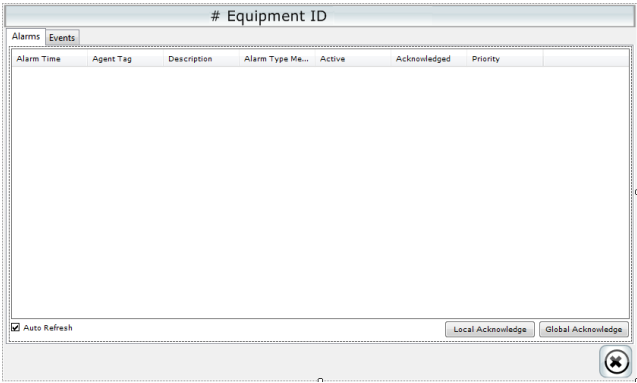
Resources	Advanced	Basic
Digital Inputs	1	1
Digital Outputs	0	0
Analog Inputs	1	1
Analog Outputs	0	0
PLC Memory Steps	441	417
System Words Used	45	45
System Bits Used	71	69
SCADA SCAN Points	18	18

Template: Advanced Flow Meter PLC Function Block

Template Graphic: Advanced Home Screen	Control	Descriptions
	Reset	Reset Flow, Hours, Daily Totals

Template Graphic: Advanced Setup Screen	Control	Descriptions
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	RawEng Min##### ##### Max##### ##### Minimum Flow Value #####.## Alarm Set points High☒#### Low☒####	Flow Rate Scaling
	Minimum Flow Value #####.##	Minimum flow rate before calculations starts
	Alarm Set points High☒#### Low☒####	Alarm Set points and enables.

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.